Dual-Branch Attention-Based Frequency Domain Network for Cross-Subject SSVEP-BCIs

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***Abstract*—Steady-state visual evoked potential-based brain-computer interfaces (SSVEP-BCIs) hold significant promise for enabling high-speed human-computer interaction in real-world scenarios. However, existing frequency-domain decoding methods treat frequency spectrum features (the real and imaginary spectrum features) as a single feature without considering their unique spatial and spectral characteristics, resulting in insufficient generalizable features and limited classification accuracy in cross-subject scenarios. To address this issue, we propose a Dual-Branch Attention-Based Frequency Domain Network (DB-AFDNet) to independently decode real and imaginary spectral components, aiming to acquire more discriminative and generalizable features for cross-subject applications. Specifically, we construct inter-branch attention similarity constraints to encourage the two branches to have similar attention properties, promoting to learn the consensus characteristics in the dual branches. Furthermore, we propose intra-branch orthogonality constraints to explore branch-specific discriminative features to learn generalizable features. Experimental studies on two public datasets, the Benchmark and Beta datasets, demonstrate that DB-AFDNet outperforms state-of-the-art methods in cross-subject classification, achieving a relative improvement of 1.36% and 1.45%, respectively.**

***Index Terms*—Steady-state visual evoked potential (SSVEP), frequency decoding, attention.**

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The code is available at https://github.com/YYingDL/DBAFDNet.

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I. Introduction

BRAIN-COMPUTER interfaces (BCIs) have received widespread attention for their ability to establish a direct communication channel between the human brain and external devices [1]. In the past few decades, electroencephalogram (EEG)-based BCIs have been applied in diverse fields, such as stroke rehabilitation [2], [3], emotion detection [4], [5], cursor control [6], [7] and neurofeedback [8], [9]. Among the various paradigms of EEG-based BCIs, steady-state visual evoked potential (SSVEP)-based BCIs have become one of the most popular paradigms because of their high signal-to-noise ratio (SNR), high information transfer rate (ITR), and low setup requirements [10], [11].

The SSVEP refers to the brain's neural response when it is subjected to visual stimulation that flashes at a specific frequency [12]. Based on this characteristic, commands in SSVEP-BCIs are encoded using stimuli with distinct flicker frequencies and phases, named joint frequency-phase modulation (JFPM) [13], [14]. When subjects turn their gaze to the character they intend to select, the decoding algorithm can decode the command selection by recognizing the SSVEP component. In the decoding process, to ensure the practicability of the SSVEP-BCIs, the SSVEP recognition algorithm must balance the recognition accuracy and the total recognition time [15].

In the past decades, various methods have been proposed to enhance the practicality of SSVEP-BCI systems. For example, researchers have developed some training-free methods,

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including power spectral density analysis (PSDA) [16], canonical correlation analysis (CCA) [17], filter bank canonical correlation analysis (FBCCA) [18]. However, the non-stationary nature of EEG signals often results in SSVEP signals consisting of substantial noise [19], [20]. These training-free approaches provide satisfactory classification accuracy only under long data lengths, which affects the ITR. To overcome this limitation, researchers have proposed several training-based frequency classification algorithms, including individual template-based CCA (IT-CCA) [21], multiset canonical correlation analysis (MsetCCA) [22], and task-related component analysis (TRCA) [23]. Nevertheless, the requirement of a calibration session for each subject is very time-consuming and can easily lead to subject fatigue [24], which significantly impacts the user experience in practical applications [25], [26]. Consequently, it is necessary to develop a cross-subject decoding method capable of accurate predictions on new subjects without requiring calibration data.

Nevertheless, significant individual differences [27] make it difficult for these traditional machine learning-based algorithms to achieve high accuracy in cross-subject classification. Recent advances in deep learning have sought to improve knowledge transfer from existing subjects to new users. For instance, several studies have demonstrated that deep learning can effectively extract rich temporal features from multi-channel EEG signals [28], [29], [30], [31]. Furthermore, capitalizing on the frequency-specific nature of SSVEP, several frequency-domain deep learning approaches have been proposed to enhance feature extraction and classification [32], [33], [34], showing promising performance in cross-subject classification scenarios. However, most current frequency-domain methods simply concatenate the real and imaginary components of the spectrum into a unified input representation [32], [35], which may overlook their inherent differences and limit feature discriminability. Although dual-branch architectures have emerged as a promising alternative, capturing complementary information across different scales [36], [37], structures [38], or domains [39], [40]. Yet, these methods often lack explicit mechanisms to enforce consistency between branches or to disentangle shared and unique feature representations, limiting their ability to generalize across subjects. As illustrated in Fig. 1, although the real and imaginary components of the spectrum share common characteristics, they also exhibit distinct spatial and spectral attributes. Conventional single-branch frequency-domain decoding approaches, however, fail to comprehensively capture these complementary characteristics, leading to insufficient feature representation and limited classification performance.

To address the aforementioned issues, this paper proposes a Dual-Branch Attention-based Frequency Domain Network (DB-AFDNet) to extract discriminative spectral features and enhance cross-subject classification performance. Unlike previous methods that focus solely on the commonalities between the real and imaginary spectral features, DB-AFDNet explicitly models both the consensus and unique representations. Specifically, we first construct the Domain Information Awareness Network (DIAN), which employs an attention mechanism to capture critical spatial and frequency information from the spectral inputs.

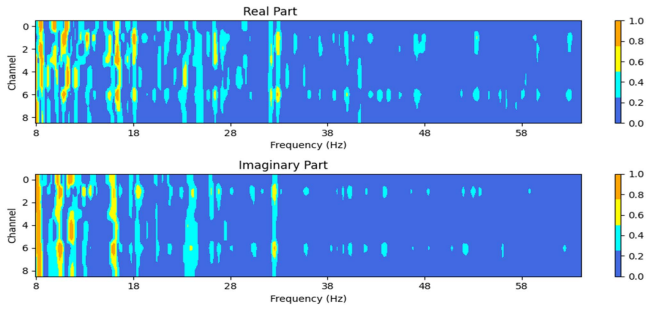


Fig. 1. The characteristic distributions of the real and imaginary parts of the spectrum features of a 1-second SSVEP signal in response to an 8 Hz target stimulus are presented. These results may indicate that these two types of features contain substantial consensus frequency information. However, some discrepancies are also observed, which may indicate unique spatial and spectral characteristics of the real and imaginary components.

Then, the real and imaginary spectral features are processed by two DIAN branches, respectively. To balance shared and unique feature learning, we introduce two novel constraints: an inter-branch attentional similarity constraint that minimizes attention divergence between branches to capture common representations, and an intra-branch orthogonality constraint that promotes disentanglement of branch-specific features. This dual-constraint strategy ensures the model learns both consensus characteristics and discriminative, subject-invariant spectral-spatial patterns. Finally, the consensus and unique attributes are fused for accurate and robust SSVEP target identification in cross-subject scenarios.

Overall, the main contributions of this paper are summarized as follows:

- To learn a more representative and generalizable feature representation, we propose a DB-AFDNet, which utilizes inter-branch attention similarity constraints and intra-branch orthogonality constraints to extract consensus and unique attributes from two frequency-domain features, respectively.
- Furthermore, we embed the attention mechanism in dual branches to explore critical spatial and spectral information in SSVEP spectrum data.
- The proposed DB-AFDNet algorithm achieves high classification accuracy on the Benchmark and Beta datasets in the cross-subject scenarios, thereby promoting the practical application of the SSVEP-BCIs paradigm in real-world scenarios.

The remainder of this paper is organized as follows: Section II describes related work. Section III introduces the proposed DB-AFDNet. Section IV presents comprehensive experimental results. Section V analyzes the effectiveness. Finally, Section VI concludes this study.

II. Related Work

A. Traditional Methods in Within-Subject Classification

The SSVEP-BCIs have received increasing attention from researchers because of their stability, high SNR, and high ITR,

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which have also driven the development of advanced decoding algorithms [6], [41]. For instance, Cheng et al. [16] proposed a PSDA approach that employed an intensity threshold based on the Fast Fourier Transform (FFT) to detect targets. Lin et al. developed CCA [17], which measured the correlation between multi-channel EEG signals and hand-crafted sinusoidal reference signals, reducing noise interference. The FBCCA method [18] extended CCA by incorporating harmonic components, thereby enhancing classification performance. Although these training-free methods are widely adopted, the non-stationarity of EEG signals makes it difficult for these methods to ensure high accuracy with limited data. This challenge has led to the development of training-based methods. For example, IT-CCA [21] and MsetCCA [22] leveraged SSVEP calibration data to derive subject-specific spatial filters, which effectively mitigated subject-specific noise interference. The TRCA method, as proposed by Nakanishi et al. [23], further improved robustness by exploring spatial filters across multiple trials to maximize event-related components. However, these traditional methods require a large amount of individual template data, which is costly and hinders the practical application of SSVEP-BCIs. Furthermore, due to significant inter-subject variability, these approaches often generalize poorly to new users in cross-subject scenarios.

B. Deep Learning for Cross-Subject Classification

The great success of deep learning methods in computer vision and natural language processing has attracted widespread attention from researchers [42]. Some researchers have employed deep learning methods for frequency recognition in cross-subject classification tasks. For example, Wattowich et al. [43] introduced a compact convolutional neural network (Compact-CNN), which automatically extracted features from raw EEG signals in the time domain. Li et al. [28] proposed convolutional correlation analysis (Conv-CA), which incorporated non-linear operations to merge EEG data from various channels and time frames and utilized a correlation layer to help constrain the fitting space. Guney et al. [29] developed a novel deep neural network (DNN) that utilized convolutions to deal with the sub-band harmonics and channels in the time domain. Zhang et al. [30] introduced a bidirectional siamese correlation analysis (bi-SiamCA) model, which employed a siamese architecture with long short-term memory (LSTM) to measure the similarity between the SSVEP signal and templates.

Furthermore, considering the characteristics of JFPM in SSVEP tasks, researchers have also explored frequency-domain deep-learning methods. For instance, Ravi et al. [32] introduced a complex spectrum features convolutional neural network (C-CNN) method, which revealed that the C-CNN method likely learned both the amplitude and phase-related information from the complex spectrum features SSVEP data for classification. Gao et al. [33] employed a zero-padding approach to enhance frequency-domain resolution after a fast Fourier transform (FFT), which enhanced signal resolution within the spectrum and thus provided abundant frequency features for classification tasks. Dang et al. [34] proposed a multi-harmonic linkage convolutional neural network (MHLCNN) that utilizes multiple branches to learn rich frequency feature representation. Chen et al. [35] introduced an FB-SSVEPformer, leveraging the transformer architecture to extract frequency-domain features and spatial features from the spectrum. Although these deep learning-based methods show promising results for cross-subject SSVEP recognition, they consider real and imaginary spectrum features as a single frequency feature without considering the unique characteristics and distributions of these two features, resulting in poor feature representation and limited performance in cross-subject classification.

C. Dual-Branch Network in EEG-Based BCIs

Recent advancements in EEG-based BCIs have witnessed the emergence of multi-branch neural networks designed to extract complementary information from different scales, structures, or domains. For instance, Tao et al. [36] introduced an attention-based dual-scale fusion convolutional neural network (ADFCNN) to simultaneously extract and fuse spectral and spatial EEG features at varying scales, enhancing motor imagery classification. Similarly, Zhou et al. [37] proposed a dual-branch attention temporal convolutional network (DB-ATCNet) to capture multi-scale, low-level local EEG patterns, improving feature diversity. On the other hand, Ye et al. [38] developed a dual-stream self-attentive adversarial graph contrastive learning framework to integrate both structural and non-structural EEG features. Besides, some studies consider extracting information from different domains (e.g., the temporal domain and the spectral domain). For example, Qin et al. [44] introduced a two-branch manifold domain enhanced transformer to holistically model temporal and spectral EEG dynamics. Additionally, Sun et al. [39] and Li et al. [40] explored dual-branch graph convolutional networks (DBGCN) and time-frequency fusion transformers (TFF-Former), respectively, to simultaneously exploit temporal and spectral EEG characteristics.

III. Methodology

Unlike existing frequency-domain deep learning decoding approaches in SSVEP-BCIs, the proposed DB-AFDNet constructs a dual-branch framework to extract rich and generalizable features from the real and imaginary spectrum features, respectively.

As illustrated in Fig. 2, the DB-AFDNet consists of two main components:

1) **Domain Information Awareness Network (DIAN)** incorporates two attention mechanisms [45], [46] to capture discriminative spatial and spectral features. The channel-wise attention mechanism learns critical spatial representations by recalibrating its multi-channel data. Meanwhile, the multi-scale adaptive attention mechanism employs convolutional kernels of varying scales to capture local and global spectral information.
2) **Representation constraint learning** aims to obtain comprehensive and generalizable frequency feature representations by capturing consensus and unique properties of the dual branches. The inter-branch attention similarity

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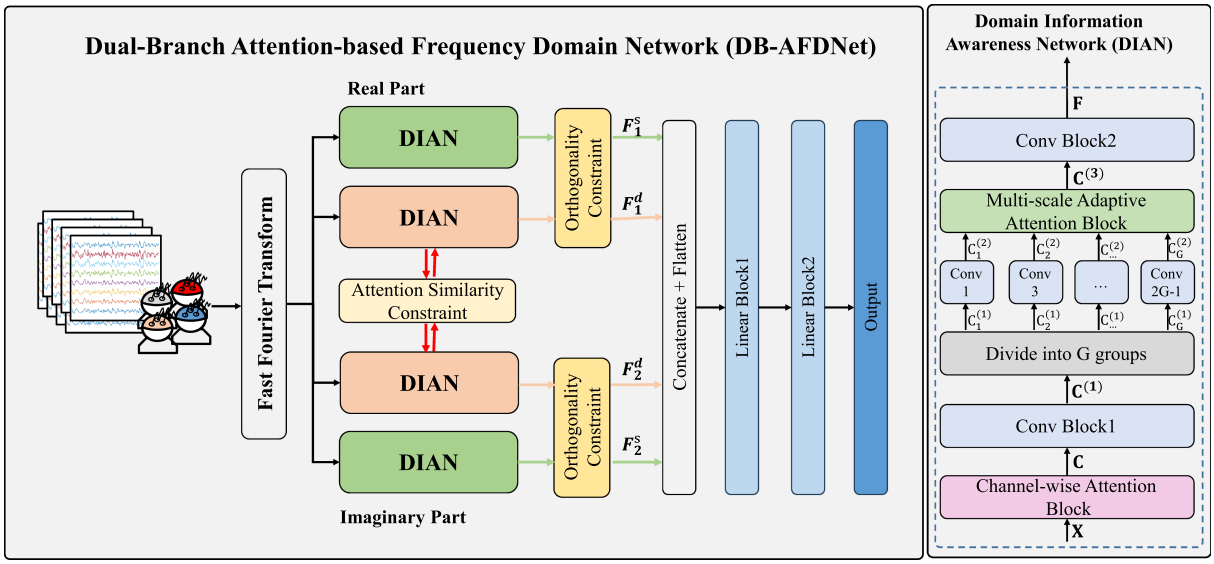


Fig. 2. A detailed illustration of the proposed DB-AFDNet. We considered two frequency-domain features as two branches and devised a DIAN to extract crucial information within each complex frequency feature independently. Specifically, we obtain the real and imaginary components of the complex frequency features using the FFT method. With the inter-branch attention similarity constraint, we obtain consensus representations $\mathbf{F}_1^{\mathrm{d}}$ and $\mathbf{F}_2^{\mathrm{d}}$. To mitigate redundancy within intra-view representations and facilitate the extraction of specific features, we introduce orthogonality constraints, resulting in specific features $\mathbf{F}_1^{\mathrm{s}}$ and $\mathbf{F}_2^{\mathrm{d}}$. Finally, we utilize concatenation methods to fuse these features for classification.

constraint aligns attention distributions between the real and imaginary branches to ensure consensus representation learning. Furthermore, an intra-branch orthogonality constraint disentangles shared and unique features, thereby reducing redundancy and promoting feature diversification within each branch.

A. Domain Information Awareness Network

1) Channel-Wise Attention: Because of the non-stability of EEG signals and interference from the external environment, multi-channel EEG signals often contain redundant or less relevant information. To improve the SNR of EEG signals, traditional machine learning methods [6], [18] usually construct spatial filters to select more relevant channels. Unlike traditional machine learning methods, we construct the channel-wise attention mechanism to recalibrate the channel relationship by learning global channel relationships [47], [48].

Specifically, $\mathbf{X} \in \mathbb{R}^{N_{ch} \times N_f}$ represents the real or imaginary part characteristics of EEG signals, where N_{ch} represents the number of channels, and N_f denotes the number of frequency features. As shown in Fig. 3, the channel-wise attention first utilizes the squeeze-and-excitation (SE) weight module [49] that embeds the global information by using the global average pooling (GAP) to obtain channel-wise statistics as $\overline{X}_{\mathrm{ch}} \in \mathbb{R}^{N_{ch}}$ as follows:

$$\overline{X}_{\mathrm{ch}} = [\overline{x}_1, \overline{x}_2, \ldots, \overline{x}_{N_{ch}}] \tag{1}$$

where $\overline{x}_j (j = 1, 2, \ldots, N_{ch})$ denotes the mean of the j-th EEG channel frequency features. Furthermore, to facilitate interaction between high and low-channel dimension information to improve the non-linearity and generalization, this channel-wise attention mechanism employs two fully connected (FC) layers.

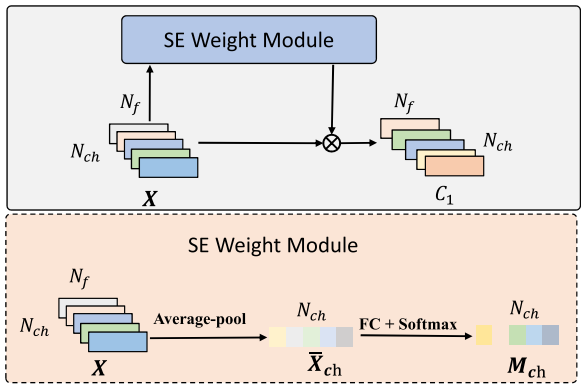


Fig. 3. A detailed illustration of the channel-wise attention block. The SE weight module first employs global average pooling for two-dimensional EEG frequency features, $\mathbf{X}$, to compress the features into a one-dimensional vector. Next, two fully connected (FC) layers enable interactions between high-dimensional and low-dimensional representations, using the softmax function to calculate an importance score. Finally, the importance score is multiplied by the original input signal to recalibrate the signal features.

Hence, we utilize the following equation to express the gating mechanism of channel-wise attention:

$$\mathbf{M}_{\mathrm{ch}} = \mathrm{softmax}(\mathbf{W_1} \cdot \sigma(\mathbf{W_0} \cdot \overline{X}_{\mathrm{ch}})) \tag{2}$$

where the softmax excitation function evaluates the importance of each channel. The function σ typically refers to the hyperbolic tangent function (tanh) and performs a linear mapping, ensuring that the output is in the range of [-1, 1]. The two FC layers contain a dimensionality-decreasing layer and a dimensionality-increasing layer, which have parameter $W_0 \in \mathbb{R}^{N_{ch} \times \frac{N_{ch}}{r}}$ and

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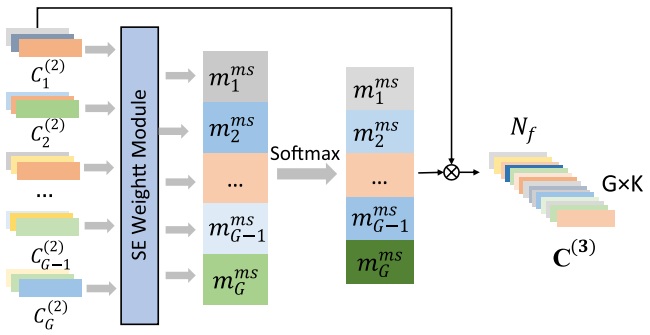


Fig. 4. A detailed illustration of a multi-scale adaptive attention block. The SE weight module is employed to obtain the importance of different scale groups, facilitating the learning of critical frequency features.

parameter $W_1 \in \mathbb{R}^{\frac{N_{ch}}{r} \times N_{ch}}$, respectively. The parameter r controls the ratio of interactions between high and low channel dimensions.

Finally, channel-wise attention weights are employed to prioritize critical channels and diminish the influence of less informative channels, thereby improving the quality and discriminative properties of the SSVEP signal. Consequently, the features $\mathbf{C}$ via the channel-wise attention can be expressed as follows:

$$\mathbf{C} = \mathbf{X} + \mathbf{X}\mathbf{M}_{\text{ch}} \tag{3}$$

2) Multi-Scale Adaptive Attention: To better capture the frequency response and corresponding harmonic components of the excitation frequency in the SSVEP signal, we employ multi-scale adaptive attention, which aims to integrate feature maps across various scales effectively [50]. As shown in Fig. 2, we use a CNN with $G \times K$ output kernels and N_{ch} input convolution kernels to extract discriminative spatial features. Here, $K(K < N_{ch})$ denotes the reduced channel dimension for computational efficiency, while G represents the number of feature groups enabling multi-scale representation learning. This design reduces parameter complexity while maintaining feature diversity.

Next, we divide $\mathbf{C^{(1)}}$ into G groups, yielding $\mathbf{C^{(1)}} = [c_1^{(1)}, c_2^{(1)}, \ldots, c_G^{(1)}]$, where each group $c_t^{(1)} = [c_{(t-1)\cdot K+1}^{(1)}, c_{(t-1)\cdot K+2}^{(1)}, \ldots, c_{t\cdot K}^{(1)}](t = 1, 2, \ldots, G)$. We then adopt the same convolution number, K, but use different convolution sizes for the t-th features $c_t^{(1)}$ to capture the frequency information. The convolutional size of the t-th features $c_t^{(1)}(t = 1, 2, \ldots, G)$ is denoted as $2t - 1$. As shown in Fig. 4, we use an SE weight module to obtain channel attention weight information from the multi-scale features $\mathbf{C^{(2)}}$:

$$\mathbf{m_t^{ms}} = \text{softmax}(\mathbf{W_3} \cdot \sigma(\mathbf{W_2} \cdot \bar{c}_t^{(2)})) \tag{4}$$

where $\bar{c}_t^{(2)}(t = 1, 2, \ldots, G)$ represents the mean of the t-th group with the kernel size is $2t - 1$. The $W_2 \in \mathbb{R}^{K \times \frac{K}{r}}$ and $W_3 \in \mathbb{R}^{\frac{K}{r} \times K}$ represent two parameters of a dimensionality-decreasing layer and a dimensionality-increasing layer in two FC layers.

Furthermore, to fuse contextual information from different scales for cross-scale interactions and generate pixel-level attention for the high-level features, we employ the concatenation method to obtain the complete multi-scale channel attention vector and apply a softmax function to acquire attention. The formula is expressed as follows:

$$\mathbf{M}_{\text{ms}} = \text{softmax}(m_1^{ms} \oplus m_2^{ms} \oplus \ldots \oplus m_G^{ms}) \tag{5}$$

where $\oplus$ represents the concatenation operator, and the $softmax$ function transforms the importance of information at different scales into probabilities. The $\mathbf{M}_{\text{ms}}$ denotes the attention obtained from different kernel sizes. Then, an element-wise multiplication is performed between the attention weight and the corresponding scale feature map:

$$\mathbf{C}^{(3)} = \mathbf{C}^{(2)} + \mathbf{C}^{(2)}\mathbf{M}_{\text{ms}} \tag{6}$$

therefore, we can obtain the different kernel scale attentive features $\mathbf{C}^{(3)}$. Then, a CNN layer is used to obtain the frequency feature $\mathbf{F}$.

B. Representation Constraint Learning

1) Inter-Branch Attention Similarity Constraint: Different from some existing methods that use distribution alignment to learn consensus attributes from dual branches, DB-AFDNet constructs inter-branch attention similarity constraints to learn consensus attributes. More specifically, we enforce the attention modules in the dual DIAN branches to possess consistent attention weights, ensuring that the two branches maintain uniform attention areas and effectively learn the crucial consensus attributes within the dual branches. The expression for the attentional similarity constraint within the attention module is given as follows:

$$\mathcal{L}_{\text{d}} = \left\|\mathbf{M}_{\text{ch}}^{\text{real}} - \mathbf{M}_{\text{ch}}^{\text{imaginary}}\right\|_F^1 + \left\|\mathbf{M}_{\text{ms}}^{\text{real}} - \mathbf{M}_{\text{ms}}^{\text{imaginary}}\right\|_F^1 \tag{7}$$

where $\mathbf{M}_{\text{ch}}^{\text{real}}, \mathbf{M}_{\text{ms}}^{\text{real}}$ represent the real spectrum attention weights of the channel-wise attention and the multi-scale adaptive attention, respectively. The $\mathbf{M}_{\text{ch}}^{\text{imaginary}}, \mathbf{M}_{\text{ms}}^{\text{imaginary}}$ represent the imaginary spectrum attention weights of the channel-wise attention and the multi-scale adaptive attention, respectively. The $|\cdot|_F^1$ represents the calculation of the L_1 distance.

2) Intra-Branch Orthogonality Constraint: The inter-branch attention similarity constraint helps capture consensus attributes between dual branches, however, it's also essential to acknowledge that each frequency feature also holds unique spatial and spectral properties that can complement the consensus attributes, leading to a more comprehensive feature representation. To effectively disentangle shared information from specific information and prevent their mutual influence, DB-AFDNet incorporates an intra-branch orthogonality constraint.

In this study, we derive the real and imaginary components of complex frequency features using the FFT technique. Next, we apply inter-branch attention similarity constraints to regulate one of the DIAN within each branch. This regulation ensures that both branches exhibit similar attention characteristics, facilitating the learning of consensus features between the two branches. We then apply an orthogonality constraint between the output of this DIAN and the output of the other DIAN, so that the latter's output corresponds to specific frequency features. Finally, this approach yields the features $(\mathbf{F}_1^{\text{s}}, \mathbf{F}_1^{\text{d}}, \mathbf{F}_2^{\text{s}}, \mathbf{F}_2^{\text{d}})$ for the

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two frequency features. Here, $\mathbf{F}_1^{\mathrm{d}}$ and $\mathbf{F}_2^{\mathrm{d}}$ represent the consensus properties of the real and imaginary components of the spectrum, and $\mathbf{F}_1^{\mathrm{s}}$ and $\mathbf{F}_2^{\mathrm{s}}$ denote the specific properties of the real and imaginary parts of the spectrum. The loss of orthogonality can be defined as follows:

$$\mathcal{L}_{\mathrm{s}} = \left\|\mathbf{F}_1^{\mathrm{d}} \cdot \mathbf{F}_1^{\mathrm{s}}\right\|_F^2 + \left\|\mathbf{F}_2^{\mathrm{d}} \cdot \mathbf{F}_2^{\mathrm{s}}\right\|_F^2 \tag{8}$$

where $\|\cdot\|_F^2$ is the squared frobenius norm.

C. Multi-Layer Decision Network

To effectively capture the consensus and unique properties of the two frequency features, the concatenation method is utilized, and then two linear layers are used for target classification. All convolutional and linear blocks in our architecture incorporate a consistent sequence of LayerNorm, GELU activation, and Dropout to enhance stability, nonlinear representation, and regularization. This design mitigates overfitting and guarantees robust feature learning and improved generalization across all modules. For the linear classification head, the first fully-connected layer comprises $6 \times N_t$ neurons, projecting high-dimensional features into an expanded space to preserve discriminative information, while the subsequent layer reduces the dimension to N_t (the number of target classes), thereby refining class-specific representations for final prediction.

Accordingly, the loss function of DB-AFDNet contains three parts, and it is formulated as the weighted sum of the classification loss $\mathcal{L}_{\mathrm{c}}$, the constraint losses ($\mathcal{L}_{\mathrm{d}} + \mathcal{L}_{\mathrm{s}}$) as:

$$\mathcal{L} = \mathcal{L}_{\mathrm{c}} + \lambda_1 \mathcal{L}_{\mathrm{d}} + \lambda_2 \mathcal{L}_{\mathrm{s}} \tag{9}$$

where λ_1, λ_2 are trade-off parameters.

IV. Experiments and Results

A. Dataset

The Benchmark dataset [14]: The Benchmark dataset comprises experimental data from 35 healthy subjects (17 females, aged 17-34 years) who participated in an SSVEP-based BCI speller experiment. The BCI speller includes 40 targets generated using the JFPM method. The frequencies range from 8 to 15.8 Hz with 0.2 Hz increments, and there is a phase difference of 0.5 π between adjacent frequencies. The experiment is divided into 6 blocks or sessions, with each block consisting of 40 randomized sequential trials, corresponding to one trial per target character. Each trial begins with a visual cue on the screen for 0.5 seconds, directing the subject's gaze to the desired target. Subsequently, a 5-second stimulus is presented, followed by a 0.5-second offset. Throughout the experiment, the subjects' EEG data were acquired using a 64-channel EEG acquisition device. The EEG signals are downsampled to a sampling rate of 250 Hz. Notably, the subjects in this dataset exhibit an estimated average visual latency of approximately 140 ms.

The Beta dataset [51]: This dataset is similar to the benchmark dataset, but there are some subtle differences. This dataset contains more subjects, i.e., 70 healthy subjects. The SSVEP-based BCI speller presents the target characters in the form of a keyboard, comprising 40 targets encoded using the JFPM method. Unlike the benchmark dataset, this experiment takes place outside of a laboratory. Consequently, the dataset had a lower SNR compared to the benchmark dataset. The experiment consists of 4 blocks, and the stimulus duration of each trial doesn't exceed 3 s. The average visual latency in this dataset is approximately 130 ms.

B. Experiment Implementation

In this experiment, subjects were instructed to fixate on each visual target in a randomized order. Meanwhile, we selected only data from nine occipital and parieto-occipital electrodes (Pz, PO5, PO3, POz, PO4, PO6, O1, Oz, and O2) for subsequent performance evaluation, similar to previous studies [10], [35], [52]. After the preprocessing stage, a total of 240 samples are obtained for each subject in the Benchmark dataset, where each sample $X_i (i = 1, 2, \ldots, 240) \in \mathbb{R}^{9\times 1250}$. For the Beta dataset, a total of 160 samples are obtained for each subject, where each sample $X_i (i = 1, 2, \ldots, 160) \in \mathbb{R}^{9\times 750}$. Then, we filter the EEG signals with a fourth-order Butterworth filter, using a band-pass range of 8–64 Hz. We apply the zero-padding fast Fourier transform (FFT) method to improve frequency resolution, which transforms the EEG signals into their corresponding frequency-domain features [33].

We conducted the experiment in a leave-one-subject-out (LOSO) manner. The data of one subject served as a test set, and the data of the remaining subjects were selected as the training set. Then, we use a data length of 1 s to capture the stable SSVEP response while achieving a balance between performance and response speed. For instance, with a data length of 1 s, for the Benchmark dataset, we used the total of training data samples $S_i (i = 1, 2, \ldots, 8160) \in \mathbb{R}^{9\times 250}$, and testing data sample $S_j (j = 1, 2, \ldots, 240) \in \mathbb{R}^{9\times 250}$. All deep learning methods are implemented using the PyTorch framework and trained on an NVIDIA RTX 2080 GPU. Specifically, these network parameters are optimized by employing the Adam optimizer to minimize the classification loss and the constraint losses. The learning rate is set to 0.001 to achieve optimal model convergence. We use L_2 regularization with a weight decay factor of 0.001 and a dropout rate of 0.5 to prevent overfitting of the model. The batch size is set to 128, and the maximum number of training epochs is limited to 200. The detailed structure of the model is listed in Table I, which includes information such as modules, layers, kernel size, number of trainable parameters, and output shape.

C. Evaluation Metrics

The machine learning decoding algorithms (FBCCA [18], TRCA [23]), time-domain deep learning decoding algorithms (EEGNet [43], Conv-CA [28], Guney-DNN [29]), frequency-domain deep learning decoding algorithms (CCNN [32], and FBSSVEPformer [35]), and dual-branch decoding algorithms (ADFCNN [36], TFF-Former [40]) are selected as baseline methods.

1) *Traditional methods:*
   a) *FBCCA:* This uses artificial sinusoidal signals as a template, with the identification result determined by the filter bank frequency that yields the maximum canonical correlation coefficient.

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TABLE I
THE PARAMETERS OF DBACFNET ARCHITECTURE

Module	Layer	Size (input size × output size × kernel size)	Params	Output Shape
FFT Processing	Real/Imaginary Component Extraction	–	-	$B \times C \times 280$
DIAN	Channel-wise attention block	$\begin{bmatrix} C \times 4 \times 1 \\ 4 \times C \times 1 \end{bmatrix}$	8C	B×C×280
	Conv block1	$[C \times 64 \times 1]$	64C	B×64×280
	Multi-scale temporal convolution	$\begin{bmatrix} 8 \times 8 \times 1 \\ \vdots \\ 8 \times 8 \times 15 \end{bmatrix}$	4,096	B×64×280
	Multi-scale adaptive attention block	$\begin{bmatrix} 8 \times 4 \times 1 \\ 4 \times 8 \times 1 \end{bmatrix} \times 8$	512	B×64×280
	Conv block2	$[64 \times 16 \times 1]$	1,024	B×16×280
Fusion	Concatenation	-	-	B×64×280
	Flatten	-	-	B×17,920
Classifier	Linear block1	$[17,920 \times 240]$	4,300,800	B×240
	Linear block2	$[240 \times N_t]$	$240N_t$	B $\times N_t$
Total Parameters	-	-	≈4.8M	
Notations	B = batch size, C = number of EEG channels	N_t = number of classes		

b) *TRCA:* This obtains the spatial filter by exploiting the reproducibility of SSVEPs across trials.

2) *Time-domain deep learning decoding methods:*
   a) *EEGNet:* This uses depthwise and separable convolutions for EEG-specific feature extraction.
   b) *Conv-CA:* This is a template-matching method based on CNN.
   c) *Guney-DNN:* This is a dense feature extraction method.
3) *Frequency-domain deep learning decoding methods:*
   a) *CCNN:* This processes complex frequency-domain data as input.
   b) *FB-SSVEPformer:* This uses a Transformer architecture for SSVEP feature extraction.
4) *Dual-branch decoding methods:*
   a) *ADFCNN:* This simultaneously extracts and fuses spectral and spatial EEG features at different scales.
   b) *TFF-Former:* This uses a Transformer architecture to simultaneously exploit temporal and spectral EEG characteristics.

For a fair comparison, we use the same training data for these methods, and we set the same hyperparameters for these deep learning methods. Besides, SSVEP signals contain harmonic components with frequencies several times greater than the fundamental frequency [18], [35]. These components can serve as classification features. Therefore, we integrate filter bank technology and propose FB-DB-AFDNet to enhance classification performance by fully leveraging harmonic information, as shown in Fig. 5.

The balanced accuracy (BA), true positive rate (TPR), false positive rate (FPR), and accuracy are used to evaluate model performance in the SSVEP task. The results are expressed as mean ± standard deviation for all test subjects. The calculation formulas are as follows:

$$\begin{cases} \text{BA} = \left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right)/2 \\ \text{TPR} = \frac{TP}{TP+FN} \\ \text{FPR} = \frac{FP}{TN+FP} \\ \text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \end{cases} \tag{10}$$

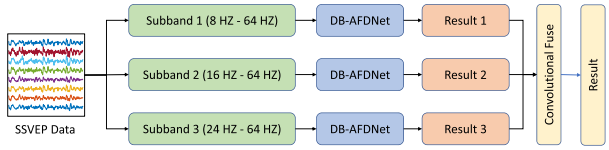


Fig. 5. A detailed illustration of FB-DB-AFDNet. We filter the original SSVEP data into three frequency subbands and feed them into the corresponding subnetworks as input. FB-DB-AFDNet uses DB-AFDNet as the backbone subnetwork, with each subnetwork processing the filtered data from a specific frequency subband. Finally, we weigh and fuse the outputs of each subnetwork to obtain the final result.

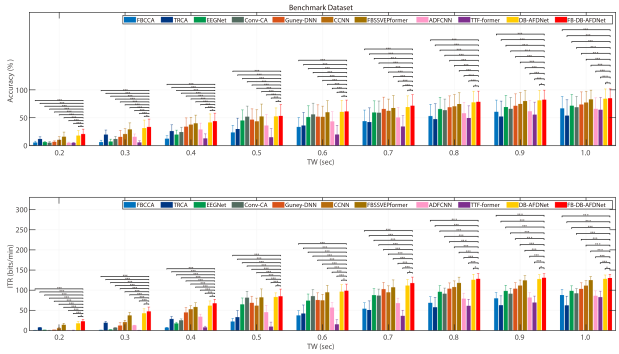


Fig. 6. The upper panel presents the mean classification accuracy, and the lower panel presents the mean information transfer rate (ITR) for all 35 subjects in the Benchmark dataset. The standard errors are indicated by the bars. Statistical significance * indicates $p < 0.05$, and *** indicates $p < 0.001$.

where TP represents the number of correctly classified positive samples, and FN represents the number of incorrectly classified positive samples. TN represents the number of correctly classified negative samples, and FP represents the number of incorrectly classified negative samples.

Besides, the ITR is also calculated to evaluate the performance of each method, which can be described as follows:

$$\text{ITR} = \frac{60}{T} \times \left[\log_2 N + P\log_2 P + (1-P)\log_2 \frac{1-P}{N-1}\right] \tag{11}$$

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TABLE II
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON TWO DATASETS WITH 1 S DATA LENGTH (MEAN ± STANDARD DEVIATION)

	Method	Benchmark dataset				Beta dataset			
		BA(%)	TPR(%)	FPR(%)	Accuracy(%)	BA(%)	TPR(%)	FPR(%)	Accuracy(%)
Classical	FBCCA	67.71 ± 21.79	77.37 ± 21.21	22.52 ± 21.18	67.92 ± 21.32	51.88 ± 22.07	50.98 ± 22.57	50.96 ± 22.11	51.85 ± 22.33
	TRCA	55.62 ± 28.47	55.82 ± 27.89	44.14 ± 27.40	55.12 ± 27.11	45.98 ± 20.50	55.89 ± 20.99	45.78 ± 20.72	45.87 ± 20.31
Time-domain	EEGNet	68.85 ± 20.42	68.35 ± 20.32	32.49 ± 20.14	67.16 ± 20.48	48.06 ± 17.44	52.46 ± 18.64	47.56 ± 18.66	47.86 ± 18.84
	Conv-CA	66.58 ± 22.33	64.89 ± 24.00	35.73 ± 22.19	66.22 ± 22.01	47.43 ± 24.35	47.85 ± 23.89	53.03 ± 22.62	48.00 ± 24.70
	Guney-DNN	72.46 ± 21.22	72.98 ± 21.22	28.65 ± 21.76	73.52 ± 21.01	57.21 ± 21.58	41.91 ± 22.66	57.94 ± 21.76	58.58 ± 22.15
Frequency-domain	CCNN	76.69 ± 19.22	75.39 ± 24.33	26.22 ± 19.96	77.08 ± 20.25	64.52 ± 22.49	63.55 ± 23.91	25.90 ± 21.65	65.25 ± 21.00
	FB-SSVEPformer	82.69 ± 17.52	82.55 ± 17.43	18.48 ± 17.22	82.68 ± 16.88	71.34 ± 18.69	71.88 ± 19.83	17.85 ± 19.66	72.98 ± 19.14
Dual-branch	ADFCNN	65.88 ± 20.21	68.81 ± 19.55	31.55 ± 20.40	65.65 ± 20.01	56.21 ± 21.33	57.77 ± 22.24	53.43 ± 22.13	55.05 ± 21.56
	TFF-Former	63.98 ± 23.21	63.45 ± 22.51	36.57 ± 22.86	64.05 ± 22.28	54.26 ± 25.37	53.25 ± 26.44	46.82 ± 27.34	54.58 ± 25.89
	DB-AFDNet	**83.41 ± 17.22**	**83.33 ± 18.36**	**16.30 ± 17.52**	**84.04 ± 17.46**	**73.85 ± 19.88**	**73.01 ± 19.77**	**26.88 ± 19.56**	**74.43 ± 19.02**
	FB-DB-AFDNet	**84.02 ± 17.21**	**84.08 ± 17.99**	**15.30 ± 17.34**	**84.83 ± 17.09**	**74.33 ± 19.04**	**73.68 ± 19.21**	**25.14 ± 19.33**	**74.65 ± 18.91**

where T is the average time, P is the classification accuracy and N represents the number of stimulus targets. Following previous studies [10], [21], [23], the time in the ITR calculation includes the gaze movement time of 0.5 s.

D. Main Results

This study used two datasets to verify the performance in cross-subject scenarios. Table II summarizes the experimental results. For the Benchmark dataset, DB-AFDNet outperformed the traditional machine learning method (FBCCA) by 16.12% in average accuracy, surpassed the time-domain deep learning method (Guney-DNN) by 10.52%, exceeded the frequency-domain deep learning method (FB-SSVEPformer) by 1.36%, and improved upon the dual-branch network (ADFCNN) by 18.39%. Similarly, on the Beta dataset, the DB-AFDNet achieved improvement results of 22.58%, 16.16%, 1.45%, and 19.38%, respectively. Furthermore, the experimental results revealed that our variant FB-DB-AFDNet, which integrates a filter bank technique, yielded additional performance gains of 0.80% and 0.22% on the Benchmark and Beta datasets, respectively. Besides, the average computation time of each data sample is 0.0062 s with a standard deviation of 0.00038 s, confirming its feasibility in practical BCI applications.

To further assess the effectiveness of DB-AFDNet, we conducted a two-tailed paired t-test to determine if the performance differences between DB-AFDNet and other methods were statistically significant. The results reveal that DB-AFDNet significantly outperforms traditional machine learning techniques, time-domain deep learning decoding methods, frequency-domain deep learning decoding methods, and dual-branch decoding methods in cross-subject scenarios (all: $p < 0.05$). Compared with other deep learning methods, our dual-branch framework effectively captures both the shared and unique characteristics of the real and imaginary spectral components, enabling the extraction of more representative and generalizable features. Furthermore, the incorporation of an attention mechanism to integrate diverse SSVEP knowledge further improves the extraction of crucial frequency features.

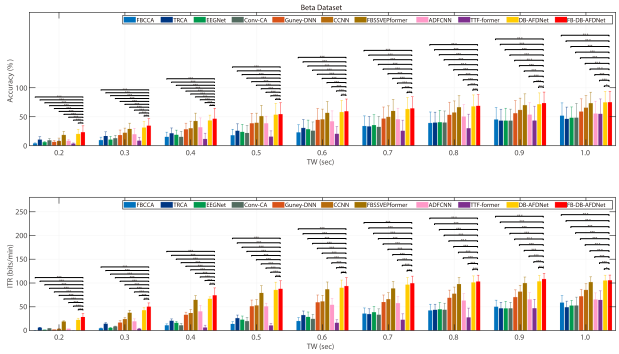


Fig. 7. The upper panel presents the mean classification accuracy, and the lower panel presents the mean information transfer rate (ITR) for all 70 subjects in the Beta dataset. The standard errors are indicated by the bars. Statistical significance * indicates $p < 0.05$, and *** indicates $p < 0.001$.

E. Performance in Different Data Lengths

To comprehensively evaluate the performance of the proposed DB-AFDNet, we conducted extensive experiments using different data lengths in two datasets. Fig. 6 and Fig. 7 illustrate the mean classification accuracy and mean ITR results of different decoding algorithms at varying data lengths (0.2 - 1 s) in the Benchmark dataset and the Beta dataset. For the Benchmark dataset, the proposed DB-AFDNet method demonstrated superior classification accuracy compared to other decoding methods when using different data lengths, with classification accuracies of 77.32%, 80.93%, and 83.42% at 0.8 s, 0.9 s, and 1.0 s, respectively. These findings indicated that longer data lengths enable SSVEP signals to contain more abundant information, thereby significantly improving the classification accuracy of the decoding algorithm. Besides, the experimental results also indicate that, at certain short data lengths (0.2 s, 0.3 s, 0.4 s, 0.5 s), our proposed DB-AFDNet method demonstrates superior generalization and classification performance than other methods.

To investigate the trade-off between real-time performance and accuracy, we calculated the results of ITR for different methods with different data lengths. The average ITR results

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TABLE III
PERFORMANCE COMPARISON OF RESULTS OBTAINED UNDER DIFFERENT INPUT DATA FORMS (SINGLE INPUT DATA AND DUAL-STREAM INPUT DATA) ON THE TWO DATASETS (MEAN ± STANDARD DEVIATION)

Dataset	Inputs	DB-AFDNet	FB-DB-AFDNet
Benchmark Dataset	$\mathbf{X}_{\text{amp}}$	54.83 ± 17.51	56.22 ± 17.01
	$\mathbf{X}_{\text{phase}}$	44.23 ± 16.88	45.32 ± 16.25
	$\mathbf{X}_{\text{real}}$	74.26 ± 22.08	75.22 ± 19.68
	$\mathbf{X}_{\text{imaginary}}$	65.53 ± 23.45	67.33 ± 22.11
	$[\mathbf{X}_{\text{amp}}, \mathbf{X}_{\text{phase}}]$	53.42 ± 18.13	54.88 ± 17.59
	$[\mathbf{X}_{\text{real}}, \mathbf{X}_{\text{imaginary}}]$	82.88 ± 17.52	83.42 ± 17.27
	$\mathbf{X}_{\text{amp}}, \mathbf{X}_{\text{phase}}$	62.18 ± 21.56	63.33 ± 22.08
	$\mathbf{X}_{\text{real}}, \mathbf{X}_{\text{imaginary}}$	**84.04 ± 17.46**	**84.83 ± 17.09**
Beta Dataset	$\mathbf{X}_{\text{amp}}$	54.58 ± 20.27	55.45 ± 19.83
	$\mathbf{X}_{\text{phase}}$	34.00 ± 13.78	34.25 ± 13.36
	$\mathbf{X}_{\text{real}}$	58.26 ± 22.58	60.12 ± 21.56
	$\mathbf{X}_{\text{imaginary}}$	55.11 ± 24.21	57.08 ± 23.56
	$[\mathbf{X}_{\text{amp}}, \mathbf{X}_{\text{phase}}]$	53.45 ± 20.29	53.82 ± 19.24
	$[\mathbf{X}_{\text{real}}, \mathbf{X}_{\text{imaginary}}]$	72.88 ±24.33	73.52 ± 22.98
	$\mathbf{X}_{\text{amp}}, \mathbf{X}_{\text{phase}}$	54.78 ± 23.09	55.43 ± 22.74
	$\mathbf{X}_{\text{real}}, \mathbf{X}_{\text{imaginary}}$	**74.43 ± 19.02**	**74.65 ± 18.91**

demonstrated that the proposed DB-AFDNet achieved the highest ITR of 154.76 bits/min at 0.9 s data length, followed by 154.65 bits/min at 0.8 s data length, and 153.52 bits/min at 1.0 s data length. Furthermore, we used the bilateral t-test statistical analysis to evaluate the effectiveness. The experimental results revealed that our proposed DB-AFDNet is superior to the other decoding methods ($p < 0.001$). Similarly, in the Beta dataset, the proposed DB-AFDNet achieved the highest accuracy compared to other methods. The classification accuracy at 0.8 s, 0.9 s, and 1 s were 67.52%, 71.02% and 74.43%, respectively. The average ITR results of the proposed DB-AFDNet method achieved a maximum ITR of 126.01 bits/min at 1 s data length, followed by 124.41 bits/min at 0.8 s data length, and 125.41 bits/min at 0.9 s data length, respectively. The results of the two-tailed paired T-test statistical analysis indicated that the proposed DB-AFDNet method exhibited significant improvements in classification performance compared to other decoding methods ($p < 0.001$). These results indicated that the proposed DB-AFDNet efficiently extracts more informative features, achieving excellent performance in cross-subject scenarios.

F. Ablation Study

To further evaluate the effectiveness of the proposed DB-AFDNet method, we conducted extensive ablation experiments. First, we conducted thorough ablation studies to analyze how different input representations affect the results. Specifically, six single-branch frequency domain features were examined, including the amplitude and phase, the real part, the imaginary part, the concatenation of amplitude and phase, and the concatenation of the real and imaginary parts of the spectrum. Then, we assessed two dual-branch representations: the amplitude and phase features of the spectrum, and the real and imaginary parts of the spectrum.

The experimental results in Table III showed that each type of frequency-domain feature contains detailed information useful for recognizing the characteristics of the SSVEP signals. In contrast, simply concatenating the features was not effective for extracting both types of features. Additionally, using the real and imaginary parts of the spectrum, instead of just calculating the amplitude and phase of spectral features, further improved classification accuracy. These results demonstrated that complex spectrum features not only can represent phase and amplitude-related features but also include more discriminative information, leading to higher classification accuracy [32], [33], [34], [35]

TABLE IV
ABLATION STUDIES ON THE TWO DATASETS (MEAN ± STANDARD DEVIATION)

Dataset	Model	BA (%)	Accuracy (%)
Benchmark Dataset	DB-AFDNet (w/o $\mathbf{M}_{\text{ch}}$)	82.71 ± 17.62	82.72 ± 17.88
	DB-AFDNet (w/o $\mathbf{M}_{\text{ms}}$)	82.97 ± 17.89	83.11 ± 17.42
	DB-AFDNet (w/o $\mathcal{L}_{\text{d}}$)	83.21 ± 18.11	83.41 ± 17.96
	DB-AFDNet (w/o $\mathcal{L}_{\text{s}}$)	82.88 ± 17.52	82.97 ± 17.27
	DB-AFDNet	**83.41 ± 17.22**	**84.04 ± 17.46**
	FB-DB-AFDNet	**84.02 ± 17.21**	**84.83 ± 17.09**
Beta Dataset	DB-AFDNet (w/o $\mathbf{M}_{\text{ch}}$)	71.88 ± 20.88	71.92 ± 20.10
	DB-AFDNet (w/o $\mathbf{M}_{\text{ms}}$)	71.96 ± 20.45	72.01 ± 19.75
	DB-AFDNet (w/o $\mathcal{L}_{\text{d}}$)	72.88 ± 20.22	72.76 ± 20.19
	DB-AFDNet (w/o $\mathcal{L}_{\text{s}}$)	71.92 ± 20.89	72.02 ± 20.34
	DB-AFDNet	**73.85 ± 19.88**	**74.43 ± 19.02**
	FB-DB-AFDNet	**74.33 ± 19.04**	**74.65 ± 18.91**

Then, we evaluate the effectiveness of each component in the proposed DB-AFDNet method. The DB-AFDNet (w/o $\mathbf{M}_{\text{ch}}$) does not employ the channel-wise attention mechanism in DIAN to extract critical spatial features for each branch. The DB-AFDNet (w/o $\mathbf{M}_{\text{ms}}$) does not utilize the multi-scale adaptive attention mechanism to capture frequency features of different scales in DIAN. The DB-AFDNet (w/o $\mathcal{L}_{\text{d}}$) does not use the attention similarity constraint in DIAN to obtain consistent features of the two frequency features. The DB-AFDNet (w/o $\mathcal{L}_{\text{s}}$) does not utilize the intra-branch orthogonality constraint to extract the view-specific features. As shown in Table IV, the ablation results on both datasets demonstrate that each module contributes positively to overall performance, and these modules are used together to achieve the best accuracy. Specifically, the attention mechanisms effectively extract discriminative spatial and spectral features by incorporating knowledge of various SSVEP characteristics. This attention similarity constraint facilitates consistent feature learning across branches, while the orthogonality constraint helps capture complementary view-specific information, collectively enriching the representation capacity.

G. Simulated Online Experiment

To further verify the effectiveness of the proposed DB-AFDNet, we collect data ourselves. We designed a 12-target simulated online SSVEP-based BCI. All subjects performed a cue-guided visual target-selecting task. In each trial, a visual cue appeared first, and the subject had to shift their gaze to the specified visual target within 1 s. After that, all visual targets started to flash simultaneously for 0.8 s. Then, the trial ended, and the next trial started. Overall, all subjects were instructed to visually select the 12 targets (frequencies: 8.0, 8.5, 9.0, $\cdots$, 13.5 Hz; phases: 0, 0.5 π, π, and 1.5 π rad) in a random order

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TABLE V
CLASSIFICATION ACCURACY OF DIFFERENT ALGORITHMS ON OUR COLLECTED DATASETS WITH 0.8 S DATA LENGTH

Subject	CCA	EEGNet	CCNN	FB-SSVEPformer	DB-AFDNet
1	33.33	64.58	39.58	54.17	**70.83**
2	18.75	29.17	27.08	41.67	**60.42**
3	25.00	25.00	27.08	16.67	**33.33**
4	47.92	**100.00**	89.58	95.83	**100.00**
5	47.92	**83.33**	75.00	77.08	79.17
6	20.83	85.42	56.25	83.33	**95.83**
7	12.50	66.67	60.42	60.42	**75.00**
8	14.58	35.42	27.08	35.42	**41.67**
9	27.08	**97.92**	**97.92**	**97.92**	**97.92**
10	54.17	**100.00**	97.92	83.33	**100.00**
11	52.08	95.83	**97.92**	89.58	95.83
12	52.08	89.58	83.33	**93.75**	91.67
Avg	33.85	72.74	64.93	69.10	**78.47**

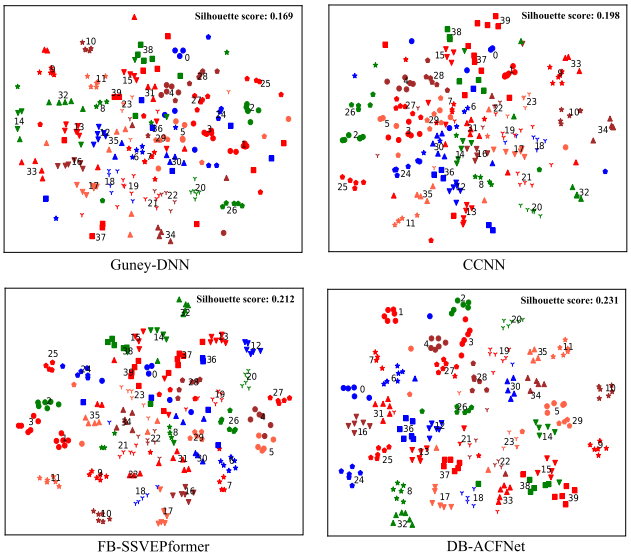


Fig. 8. The t-SNE visualization results of a representative subject with the four models. Each dot represents sample data, and different colors and shapes indicate different categories.

four times, resulting in 48 trials. Then, we used the leave-one-subject-out strategy to conduct experiments on this dataset to verify the effectiveness of different decoding methods.

Table V presents the classification accuracy for each subject in our independent validation dataset, comparing DB-AFDNet against baseline methods under identical 0.8 s window conditions. This result demonstrates that our proposed DB-AFDNet outperforms the other methods (CCA, EEGNet, CCNN, FB-SSVEPformer), which proves that our proposed DB-AFDNet can extract richer generalization features and thus significantly improve the classification performance.

H. Visualization

To better understand the effectiveness of our proposed DB-AFDNet, we utilize the t-SNE visualization to evaluate the effectiveness of feature extraction. Specifically, we select subject 15 as the test subject in the Benchmark datasets when using different decoding algorithms.

As shown in Fig. 8, this result indicates that the features extracted by our proposed DB-AFDNet exhibit a small intra-class distance and a large inter-class distance compared to Guney-DNN, CCNN, and FB-SSVEPformer. This finding suggests that our proposed method can extract richer generalization features, leading to improved classification performance. Besides, to further quantify the distribution differences of different algorithms, we employ the silhouette coefficient [53] to evaluate clustering performance. This metric assesses cluster compactness and separation by computing the ratio of the mean intra-cluster distance to the mean nearest-cluster distance for each sample, yielding scores that range from -1 (worst) to 1 (optimal). Experimental results demonstrate that our proposed DB-AFDNet method achieves a silhouette score of 0.231, which is significantly higher than those of the baseline algorithms, 0.169 for Guney-DNN, 0.198 for CCNN, and 0.212 for FB-SSVEPformer. This improvement indicates that DB-AFDNet extracts more generalizable and discriminative features, reflecting its superior clustering ability.

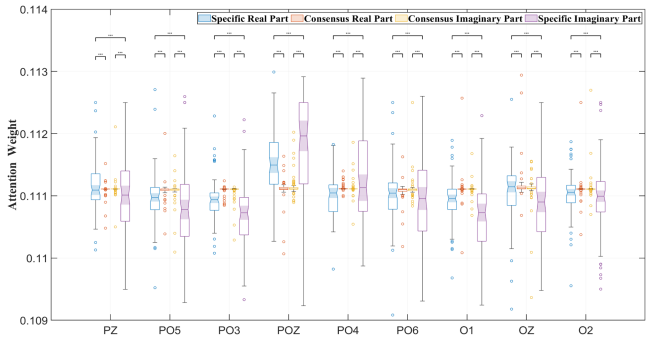


Fig. 9. Visualization of channel-wise attention weights for 70 subjects from the Beta dataset. Statistical significance is indicated by ***, $p < 0.001$. The results were obtained when the subjects were the test subjects.

V. DISCUSSION

This study employs channel-wise and multi-scale adaptive attention mechanisms to capture spatial and spectral features of different scales. To further investigate these characteristics, the attention weight visualization method is used to show the attention scores of different subjects. Specifically, since the Beta dataset has more subjects (70 subjects), the attention visualization results for all subjects on this dataset are shown when they are in the test subject.

Fig. 9 illustrates the weights of the channel attention mechanisms after applying the inter-branch attention similarity constraint and intra-branch orthogonality constraint. The results indicate that the inter-branch attention similarity constraint effectively enforces consistent attentional weighting across the dual branches. Besides, the statistical analyses demonstrate that the attention weights derived from the intra-branch orthogonality constraints significantly differed from those associated with the consensus features ($p < 0.001$). Moreover, the attention weights of specific features between the two branches exhibited significant dissimilarities ($p < 0.001$). For the attention weight of consensus features, the channel attention weights reveal that PZ, POZ, PO4, OZ, and O2 have higher average attention weights, suggesting that these electrodes correspond to brain regions

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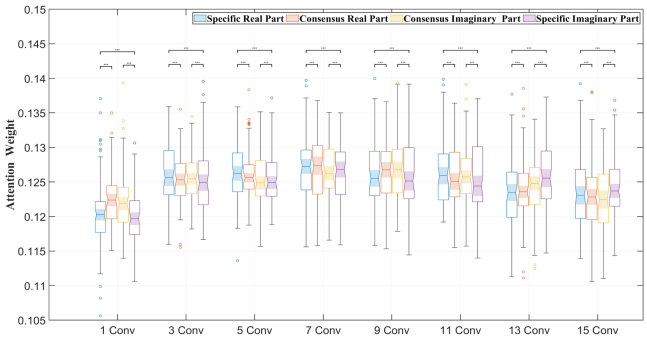


Fig. 10. Visualization of multi-scale adaptive attention weights for 70 subjects from the Beta dataset. Statistical significance is indicated by ***, $p < 0.001$. The results were obtained when the subjects were the test subjects.

more responsive to visual stimulation. Some previous studies have also demonstrated that the occipital lobe region is closely related to visual stimulus arousal [54], [55], [56]. Regarding the attentional weighting of specific features, we observed that the real part of the spectrum exhibited higher weights than the imaginary part in the left occipital region (PO5, PO3, O1, OZ). Additionally, we analyzed the frequency feature representations across different scales. Furthermore, specific attention to both the real and imaginary parts of the spectrum exhibited higher weights than consensus attention at the POZ electrode, whereas attention was lower at the PO3, PO5, and O1 electrodes.

Fig. 10 shows the attention weights for different convolution kernel sizes corresponding to different spectral receptive fields of the multi-scale adaptive attention. These results demonstrate that convolutions of different sizes have varying degrees of importance to the model. The T-paired statistical test also reveals a significant difference between the attention weight assigned to consensus features and the weight assigned to specific features. Specifically, the multi-scale adaptive attention model assigns higher weights to convolution kernel sizes of 5 and 3 compared to a kernel size of 1. These results indicate that frequency features near the stimulus frequency contribute to richer feature extraction for classification. The convolution kernels with sizes 3 and 5 and the convolution kernels with sizes 9 and 11 exhibit higher average attention weight values, suggesting that these convolution kernels contribute to information extraction. Previous studies [45], [46] have indicated that smaller convolution kernels tend to focus on local information, whereas larger convolution kernels capture global information. Furthermore, the specific attention to the real part of the spectrum carries a higher weight than the specific attention to the imaginary part of the spectrum at small convolutional scales, whereas, at large scales, it exhibits a lower weight. This result suggests that real part frequency features elicit a stronger SSVEP response than imaginary part frequency features.

Although SSVEP-BCIs have a high ITR, they may cause visual fatigue or changes in their cognitive load. This is a recognized limitation in the current design of SSVEP-BCIs, as noted in previous studies [24], [57]. Addressing this challenge remains an important area for future research. Besides, online experiments are crucial for SSVEP-BCI systems, and it is necessary to validate the effectiveness of the algorithm in online systems in future work.

VI. Conclusion

This paper proposes DB-AFDNet to further learn rich representative and generalizable features to improve the accuracy in cross-subject classification. Specifically, unlike previous methods that concatenate the real and imaginary parts of the spectrum features, DB-AFDNet constructs two DIAN branches to extract rich features representing brain activity in two frequency domain features, respectively. In the DIAN, the attention mechanism embeds the knowledge of multi-channel SSVEP characteristics to help effectively extract critical spatial and spectral features. Then, an inter-branch attention similarity constraint is developed to learn the consensus properties of two frequency features. Furthermore, to capture unique properties that complement the consensus properties to obtain richer frequency features, an intra-branch orthogonal constraint is constructed to extract specific properties in each branch. Finally, to validate the effectiveness of the proposed algorithm, we conducted extensive experiments on the Benchmark and Beta datasets. Experimental results demonstrate that the proposed DB-AFDNet method achieves state-of-the-art performance and significantly outperforms existing methods. Ablation experiments and visualization analysis indicate that the three strategies we proposed are beneficial to improving classification performance, and the combination of the three strategies can achieve the best performance. In addition, the proposed DB-AFDNet algorithm not only has superior performance but also focuses on the expression of features, promoting further the application of the SSVEP-BCI system.

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